Roll No.: DL.AI.U4AID24120

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Amrita School of Artificial Intelligence, Faridabad

B.Tech., Mid-Term Examinations – October 2025

Third Semester

Artificial Intelligence & Data Science

23AID202 Introduction to Robotics

Duration: Two hours | Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Explain facts pertaining to robotics including history, sub-fields, and applications.
CO02	Explain the elementary concepts required for modelling robotic systems.
CO03	Develop mathematical and mechanistic models for simple robotic systems.
CO04	Use tools such as Robotics toolbox to program and visualize simple robotic systems

Answer all questions

PART I

1. The term "robot" was derived from a word meaning —
A) Work B) Slave labor C) Machine D) Automation
[1][CO01][BTL Level: 1 – Remember]

2. Robots are used in manufacturing industries mainly for —
A) Painting, welding, and assembly B) Cooking C) Data processing D) Chemical mixing
[1][CO01][BTL Level: 2 – Understand]

3. The earliest robots were primarily developed for —
A) Industrial automation B) Education C) Medical applications D) Agriculture
[1][CO01][BTL Level: 1 – Remember]

4. Collaborative robots (cobots) are designed to —
A) Work safely with humans B) Replace humans C) Perform underwater tasks D) Be teleoperated
[1][CO01][BTL Level: 2 – Understand]

5. A SCARA robot is primarily used for —
A) Pick-and-place operations B) Painting C) Excavation D) Welding
[1][CO01][BTL Level: 1 – Remember]

6. Robots used for bomb disposal and surveillance are categorized as —
A) Defense robots B) Industrial robots C) Domestic robots D) Space robots
[1][CO01][BTL Level: 2 – Understand]

7. The 3D printing robots are examples of —
A) Manufacturing automation B) Agricultural robotics C) Domestic robots D) Underwater robotics
[1][CO01][BTL Level: 3 – Apply]

8. The main advantage of using robots in hazardous environments is —
A) Safety of humans B) Increased cost C) Reduced accuracy D) High manpower
[1][CO01][BTL Level: 2 – Understand]

9. The field of robotics integrates —
A) Mechanical, Electrical, and Computer Engineering B) Civil and Chemical Engineering C) Physics and Art D) Only Programming
[1][CO01][BTL Level: 1 – Remember]

10. Humanoid robots are designed to —
A) Mimic human structure and motion B) Work underwater C) Fly autonomously D) Produce sound
[1][CO01][BTL Level: 2 – Understand]

11. Kinematics in robotics deals with —
A) Motion without considering forces B) Power flow C) Friction losses D) Motor torque
[1][CO02][BTL Level: 2 – Understand]

12. The end-effector of a robot is used to —
A) Interact with the environment B) Measure current C) Transfer data D) Store memory
[1][CO02][BTL Level: 1 – Remember]

13. The total reachable area of a robot manipulator is known as —
A) Workspace B) Plane of motion C) Orientation field D) Volume of motion
[1][CO02][BTL Level: 2 – Understand]

14. The degree of freedom (DOF) in a manipulator represents —
A) Number of independent movements B) Number of links C) Torque capacity D) Force control
[1][CO02][BTL Level: 2 – Understand]

15. The Denavit–Hartenberg (D–H) parameters describe —
A) Geometry between successive links B) Power supply C) Velocity profile D) Thermal effects
[1][CO02][BTL Level: 3 – Apply]

16. Inverse kinematics finds —
A) Joint angles for desired end-effector position B) Link length C) End-effector velocity D) Force torque
[1][CO02][BTL Level: 3 – Apply]

17. The homogeneous transformation matrix combines —
A) Rotation and translation B) Scaling C) Shearing D) Reflection
[1][CO02][BTL Level: 2 – Understand]

18. The Jacobian matrix relates —
A) Joint velocities to end-effector velocities B) Power to torque C) Temperature to speed D) Weight to distance
[1][CO02][BTL Level: 3 – Apply]

19. A revolute joint allows —
A) Rotary motion B) Linear motion C) Scaling D) Frictionless rotation
[1][CO02][BTL Level: 1 – Remember]

20. The transformation between two frames is represented as —
A) T_AB B) J_AB C) V_AB D) D_AB
[1][CO02][BTL Level: 3 – Apply]

21. The dot product of two vectors gives a —
A) Scalar B) Vector C) Tensor D) Matrix
[1][CO03][BTL Level: 1 – Remember]

22. The cross product of two vectors results in —
A) A vector perpendicular to both B) A scalar C) A matrix D) A tensor
[1][CO03][BTL Level: 2 – Understand]

23. The determinant of a rotation matrix is —
A) +1 B) −1 C) 0 D) Variable
[1][CO03][BTL Level: 2 – Understand]

24. A 4×4 homogeneous transformation matrix is used to represent —
A) Rotation and translation in 3D B) Only rotation C) Scaling only D) Projection
[1][CO03][BTL Level: 3 – Apply]

25. The last row of a homogeneous transformation matrix is always —
A) [0 0 0 1] B) [1 0 0 1] C) [0 1 0 1] D) [1 1 1 1]
[1][CO03][BTL Level: 1 – Remember]

26. The rotation about the Z-axis by θ is represented as —
A) [[cosθ, −sinθ, 0],[sinθ, cosθ, 0],[0,0,1]] B) Identity C) [[1,0,0],[sinθ, cosθ, 0],[cosθ, sinθ, 0]] D) Zero matrix
[1][CO03][BTL Level: 3 – Apply]

27. Homogeneous coordinates represent a point as —
A) $[x\ y\ z\ 1]^T$ B) [x y z] C) [x y] D) [x y z 0]
[1][CO03][BTL Level: 2 – Understand]

28. The magnitude of a vector A = (x, y, z) is —
A) $\sqrt{(x^2 + y^2 + z^2)}$ B) x + y + z C) $x^2 + y^2 + z^2$ D) 1
[1][CO03][BTL Level: 2 – Understand]

29. The inverse of a rotation matrix is equal to its —
A) Transpose B) Determinant C) Negative D) Identity
[1][CO03][BTL Level: 3 – Apply]

30. The product of two homogeneous transformation matrices gives —
A) Combined motion of the robot B) Zero C) Inverse D) Random value
[1][CO03][BTL Level: 3 – Apply]

PART II

31. What is the significance of a homogeneous transformation matrix?
[5][CO03][BTL Level: 3 – Apply]

32. Explain the importance of composition of motions in robotic kinematics.
[5][CO02][BTL Level: 4 – Analyze]

PART III

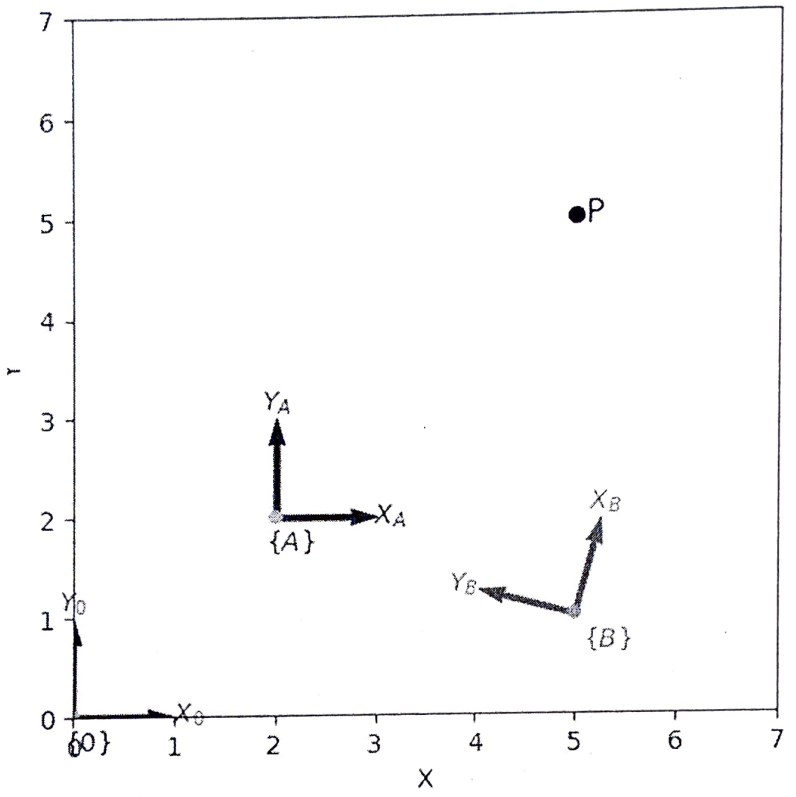
33. The **figure below** shows three coordinate frames: the base frame **{0}**, frame **{A}** translated by (2, 2), and frame **{B}** translated by (5, 1) and rotated by 75°.
A point **P(5, 5)** is represented in the world coordinate frame.

i) Using the given transformations, compute the coordinates of **P** with respect to frame **{A}** using the homogeneous transformation matrices

ii) Using the given transformations, compute the coordinates of **P** with respect to frame **{B}** using the homogeneous transformation matrices.

(Show all the necessary steps and matrix calculations.)

[10][CO03][BTL Level: 4 – Analyze]



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CO	Marks	BTL	Marks
CO01	10	BTL 1	8
CO02	15	BTL 2	13
CO03	25	BTL 3	14
CO04		BTL 4	15
CO05		BTL 5	
CO06		BTL 6	

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table